MA385: Tutorial Sheet 5

This sheet is for Week 9 tutorial. It is quite short because there are outstanding questions on the previous sheet, and also, because of a public holiday and class test, we have not covered too much new material since last week.

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Q1. Every step of Gaussian Elimination can be thought of as a left multiplication by a unit lower triangular matrix. That is, we obtain an upper triangular matrix U by multiplying A by k unit lower triangular matrices: $L_k L_{k-1} L_{k-2} \dots L_2 L_1 A = U$, where each $L_i = I + \mu_{pq} E^{(pq)}$, and $E^{(pq)}$ is the matrix whose only non-zero entry is $e_{pq} = 1$. Give an expression for k in terms of n.

Q2. Let $L = (l_{ij})$ be a lower triangular (LT) matrix of size $n \times n$. Show that $\det(L) = \prod_{j=1}^{n} l_{jj}$. Similarly, show that if $U = (u_{ij})$ is an upper triangular matrix, then $\det(U) = \prod_{j=1}^{n} u_{jj}$.

Q3. The Cauchy-Binet formula for determinants tells us that, if A and B are square matrices of the same size, then $\det(AB) = \det(A)\det(B)$. Using it, or otherwise, show that $\det(\sigma A) = \sigma^n \det(A)$ for any $A \in \mathbb{R}^{n \times n}$ and any scalar $\sigma \in \mathbb{R}$.